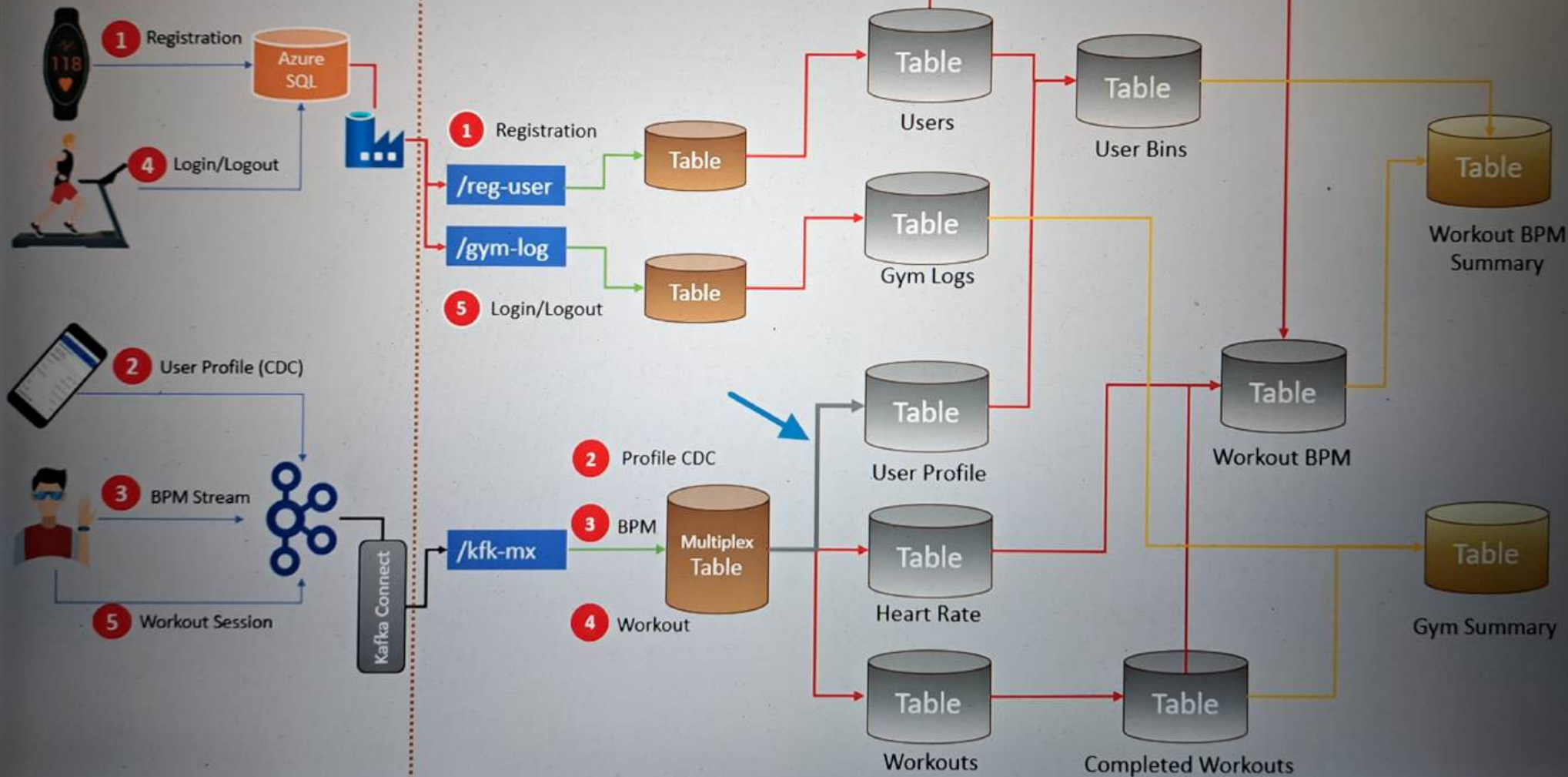






id	first name	last name	gender	dob	address	city	state	zip
1	prashant	pandey	M	15-03-1985	xyz street	bangalore	karnataka	836129

id	first name	last name	gender	dob	address	city	state	zip	Updated
1	Prashant	Pandey	M	15-03-1985	xyz street	bangalore	karnataka	836129	19-08-2023 10.00
1	Prashant	Pandey	M	15-03-1985	F1, Delta Street	bangalore	karnataka	836129	19-08-2023 10.06
1	Prashant	Pandey	M	15-04-1982	xyz street	bangalore	karnataka	836129	19-08-2023 10.02



id	first name	last name	gender	dob	address	city	state	zip
1	Prashant	Pandey	M	15-04-1982	xyz street	bangalore	karnataka	836129

id	first name	last name	gender	dob	address	city	state	zip	Updated
1	Prashant	Pandey	M	15-03-1985	xyz street	bangalore	karnataka	836129	19-08-2023 10.00
1	Prashant	Pandey	M	15-03-1985	F1, Delta Street	bangalore	karnataka	836129	19-08-2023 10.06
1	Prashant	Pandey	M	15-04-1982	xyz street	bangalore	karnataka	836129	19-08-2023 10.02



id	first name	last name	gender	dob	address	city	state	zip
1	prashant	pandey	M	15-03-1985	xyz street	bangalore	karnataka	836129

id	first name	last name	gender	dob	address	city	state	zip	Updated	Rank-Desc
1	Prashant	Pandey	M	15-03-1985	F1, Delta Street	bangalore	karnataka	836129	19-08-2023 10.06	1
1	Prashant	Pandey	M	15-04-1982	xyz street	bangalore	karnataka	836129	19-08-2023 10.02	2
1	Prashant	Pandey	M	15-03-1985	xyz street	bangalore	karnataka	836129	19-08-2023 10.00	3

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Cmd 1

```
1 class Config():
2     def __init__(self):
3         self.base_dir_data = spark.sql("describe external location `data_zone`").select("url").collect()[0][0]
4         self.base_dir_checkpoint = spark.sql("describe external location `checkpoint`").select("url").collect()[0][0]
5         self.db_name = "sbit_db"
6         self.maxFilesPerTrigger = 1000
```

Shift+Enter to run

Shift+Ctrl+Enter to run selected text

01-config - Databricks

adb-3007342644227763.3.azure.databricks.net/?o=3007342644227763#notebook/4220298042636154/command/4220298042636155

Microsoft Azure databricks Search data, notebooks, recents, and more... CTRL + P

sbitt-ws prashant@scholarnest.com

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Cmd 1

```
1 class Config():
2     def __init__(self):
3         self.base_dir_data = spark.sql("describe external location `data_zone`").select("url").collect()[0][0]
4         self.base_dir_checkpoint = spark.sql("describe external location `checkpoint`").select("url").collect()[0][0]
5         self.db_name = "sbit_db"
6         self.maxFilesPerTrigger = 1000
```

Shift+Enter to run
Shift+Ctrl+Enter to run selected text

→ 1. Centralize your configurations in one place.
2. Avoid hardcoding values.

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Cmd 1

```
1 %run ./01-config
```

Cmd 2

```
1 class SetupHelper():
2     def __init__(self, env):
3         Conf = Config()
4         self.landing_zone = Conf.base_dir_data + "/raw"
5         self.checkpoint_base = Conf.base_dir_checkpoint + "/checkpoints"
6         self.catalog = env
7         self.db_name = Conf.db_name
8         self.initialized = False
9
10    def create_db(self):
11        spark.catalog.clearCache()
12        print(f"Creating the database {self.catalog}.{self.db_name}...", end='')
13        spark.sql(f"CREATE DATABASE IF NOT EXISTS {self.catalog}.{self.db_name}")
14        spark.sql(f"USE {self.catalog}.{self.db_name}")
15        self.initialized = True
16        print("Done")
17
18    def create_registered_users_bz(self):
19        if(self.initialized):
20            print(f"Creating registered_users_bz table...", end='')
```


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print("Done")

```
def create_registered_users_bz(self):
    if(self.initialized):
        print(f"Creating registered_users_bz table...", end='')
        spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.registered_users_bz(
            user_id long,
            device_id long,
            mac_address string,
            registration_timestamp double,
            load_time timestamp,
            source_file string
        )
        """)
        print("Done")
    else:
        raise ReferenceError("Application database is not defined. Cannot create table in default database.")
```

```
def create_gym_logins_bz(self):
    if(self.initialized):
        print(f"Creating gym_logins_bz table...", end='')
        spark.sql(f"""CREATE OR REPLACE TABLE {self.catalog}.{self.db_name}.gym_logins_bz(
            mac_address string,
            gym bigint,
            login double,
```


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```
35 def create_gym_logins_bz(self):
36     if(self.initialized):
37         print(f"Creating gym_logins_bz table...", end='')
38         spark.sql(f"""CREATE OR REPLACE TABLE {self.catalog}.{self.db_name}.gym_logins_bz(
39             mac_address string,
40             gym bigint,
41             login double,
42             logout double,
43             load_time timestamp,
44             source_file string
45         )
46         """)
47         print("Done")
48     else:
49         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
50
51
52 def create_kafka_multiplex_bz(self):
53     if(self.initialized):
54         print(f"Creating kafka_multiplex_bz table...", end='')
55         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.kafka_multiplex_bz(
56             key string,
57             value string,
58             topic string,
59             partition bigint,
60             offset bigint,
```

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```
50
51
52 def create_kafka_multiplex_bz(self):
53     if(self.initialized):
54         print(f"Creating kafka_multiplex_bz table...", end='')
55         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.kafka_multiplex_bz(
56             key string,
57             value string,
58             topic string,
59             partition bigint,
60             offset bigint,
61             timestamp bigint,
62             date date,
63             week_part string,
64             load_time timestamp,
65             source_file string)
66             PARTITIONED BY (topic, week_part)
67             """)
68         print("Done")
69     else:
70         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
71
72
73 def create_users(self):
74     if(self.initialized):
75         print(f"Creating users table...", end='')
76         spark.sql(f"""CREATE OR REPLACE TABLE {self.catalog}.{self.db_name}.users(
```

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```
71
72
73 def create_users(self):
74     if(self.initialized):
75         print(f"Creating users table...", end='')
76         spark.sql(f"""CREATE OR REPLACE TABLE {self.catalog}.{self.db_name}.users(
77             user_id bigint,
78             device_id bigint,
79             mac_address string,
80             registration_timestamp timestamp
81         )
82         """)
83         print("Done")
84     else:
85         raise ReferenceError("Application database is not defined. Cannot create table in default database.
86 ")
87
88 def create_gym_logs(self):
89     if(self.initialized):
90         print(f"Creating gym_logs table...", end='')
91         spark.sql(f"""CREATE OR REPLACE TABLE {self.catalog}.{self.db_name}.gym_logs(
92             mac_address string,
93             gym bigint,
94             login timestamp,
95             logout timestamp
96         )
97         """)
```


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")

def create_gym_logs(self):

if(self.initialized):

print(f"Creating gym_logs table...", end='')

spark.sql(f"""CREATE OR REPLACE TABLE {self.catalog}.{self.db_name}.gym_logs(

mac_address string,

gym bigint,

login timestamp,

logout timestamp

)

""")

print("Done")

else:

raise ReferenceError("Application database is not defined. Cannot create table in default database.")

def create_user_profile(self):

if(self.initialized):

print(f"Creating user_profile table...", end='')

spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.user_profile(

user_id bigint,

dob DATE,

sex STRING,

gender STRING,

first_name STRING,

last_name STRING,

street address STRING.

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```
100
101 def create_user_profile(self):
102     if(self.initialized):
103         print(f"Creating user_profile table...", end='')
104         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.user_profile(
105             user_id bigint,
106             dob DATE,
107             sex STRING,
108             gender STRING,
109             first_name STRING,
110             last_name STRING,
111             street_address STRING,
112             city STRING,
113             state STRING,
114             zip INT,
115             updated TIMESTAMP)
116             """)
117         print("Done")
118     else:
119         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
120
121 def create_heart_rate(self):
122     if(self.initialized):
123         print(f"Creating heart_rate table...", end='')
124         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.heart_rate(
125             device_id LONG,
126             time TIMESTAMP,
```

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```
119         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
120
121     def create_heart_rate(self):
122         if(self.initialized):
123             print(f"Creating heart_rate table...", end='')
124             spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.heart_rate(
125                 device_id LONG,
126                 time TIMESTAMP,
127                 heartrate DOUBLE,
128                 valid BOOLEAN)
129             """)
130             print("Done")
131         else:
132             raise ReferenceError("Application database is not defined. Cannot create table in default database.")
133
134
135     def create_user_bins(self):
136         if(self.initialized):
137             print(f"Creating user_bins table...", end='')
138             spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.user_bins(
139                 user_id BIGINT,
140                 age STRING,
141                 gender STRING,
142                 city STRING,
143                 state STRING)
144             """)
145             print("Done")
```

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```
133
134
135 def create_user_bins(self):
136     if(self.initialized):
137         print(f"Creating user_bins table...", end='')
138         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.user_bins(
139             user_id BIGINT,
140             age STRING,
141             gender STRING,
142             city STRING,
143             state STRING)
144             """)
145         print("Done")
146     else:
147         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
148
149
```

```
150
151 def create_workouts(self):
152     if(self.initialized):
153         print(f"Creating workouts table...", end='')
154         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.workouts(
155             user_id INT,
156             workout_id INT,
157             time TIMESTAMP,
158             action STRING,
159             session_id INT)
160             """)
```


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```
148
149
150 def create_workouts(self):
151     if(self.initialized):
152         print(f"Creating workouts table...", end='')
153         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.workouts(
154             user_id INT,
155             workout_id INT,
156             time TIMESTAMP,
157             action STRING,
158             session_id INT)
159             """)
160         print("Done")
161     else:
162         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
163
164
165 def create_completed_workouts(self):
166     if(self.initialized):
167         print(f"Creating completed_workouts table...", end='')
168         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.completed_workouts(
169             user_id INT,
170             workout_id INT,
171             session_id INT,
172             start_time TIMESTAMP,
173             end_time TIMESTAMP)
174             """)
```

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203

```
def create_workout_bpm(self):
    if(self.initialized):
        print(f"Creating workout_bpm table...", end='')
        spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.workout_bpm(
            user_id INT,
            workout_id INT,
            session_id INT,
            start_time TIMESTAMP,
            end_time TIMESTAMP,
            time TIMESTAMP,
            heartrate DOUBLE)
        """)
        print("Done")
    else:
        raise ReferenceError("Application database is not defined. Cannot create table in default database.")
```

```
def create_date_lookup(self):
    if(self.initialized):
        print(f"Creating date_lookup table...", end='')
        spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.date_lookup(
            date date,
            week int,
            year int,
```

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```
195
196
197 def create_date_lookup(self):
198     if(self.initialized):
199         print(f"Creating date_lookup table...", end='')
200         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.date_lookup(
201             date date,
202             week int,
203             year int,
204             month int,
205             dayofweek int,
206             dayofmonth int,
207             dayofyear int,
208             week_part string)
209         """)
210         print("Done")
211     else:
212         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
213
214 def create_workout_bpm_summary(self):
215     if(self.initialized):
216         print(f"Creating workout_bpm_summary table...", end='')
217         spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.workout_bpm_summary(
218             workout_id INT,
219             session_id INT,
220             user_id BIGINT,
221             age STRING,
```


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```
210         print("Done")
211     else:
212         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
213
214     def create_workout_bpm_summary(self):
215         if(self.initialized):
216             print(f"Creating workout_bpm_summary table...", end='')
217             spark.sql(f"""CREATE TABLE IF NOT EXISTS {self.catalog}.{self.db_name}.workout_bpm_summary(
218                 workout_id INT,
219                 session_id INT,
220                 user_id BIGINT,
221                 age STRING,
222                 gender STRING,
223                 city STRING,
224                 state STRING,
225                 min_bpm DOUBLE,
226                 avg_bpm DOUBLE,
227                 max_bpm DOUBLE,
228                 num_recordings BIGINT)
229             """)
230             print("Done")
231         else:
232             raise ReferenceError("Application database is not defined. Cannot create table in default database.")
233
234     def create_gym_summary(self):
235         if(self.initialized):
236             print(f"Creating gym_summary table...", end='')
```

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```
231         else:
232             raise ReferenceError("Application database is not defined. Cannot create table in default database.")
233
234     def create_gym_summary(self):
235         if(self.initialized):
236             print(f"Creating gym_summar gold view...", end='')
237             spark.sql(f"""CREATE OR REPLACE VIEW {self.catalog}.{self.db_name}.gym_summary AS
238
239                 SELECT to_date(login::timestamp) date,
240                       gym, l.mac_address, workout_id, session_id,
241                       round((logout::long - login::long)/60,2) minutes_in_gym,
242                       round((end_time::long - start_time::long)/60,2) minutes_exercising
243                 FROM gym_logs l
244                 JOIN (
245                     SELECT mac_address, workout_id, session_id, start_time, end_time
246                     FROM completed_workouts w INNER JOIN users u ON w.user_id = u.user_id) w
247                 ON l.mac_address = w.mac_address
248                 AND w.start_time BETWEEN l.login AND l.logout
249                 order by date, gym, l.mac_address, session_id
250             """)
251             print("Done")
252         else:
253             raise ReferenceError("Application database is not defined. Cannot create table in default database.")
254
255     def setup(self):
256         import time
257         start = int(time.time())
258         print(f"\nStarting setup...")
```

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```
252         raise ReferenceError("Application database is not defined. Cannot create table in default database.")
253
254     def setup(self):
255         import time
256         start = int(time.time())
257         print(f"\nStarting setup ...")
258         self.create_db()
259         self.create_registered_users_bz()
260         self.create_gym_logins_bz()
261         self.create_kafka_multiplex_bz()
262         self.create_users()
263         self.create_gym_logs()
264         self.create_user_profile()
265         self.create_heart_rate()
266         self.create_workouts()
267         self.create_completed_workouts()
268         self.create_workout_bpm()
269         self.create_user_bins()
270         self.create_date_lookup()
271         self.create_workout_bpm_summary()
272         self.create_gym_summary()
273         print(f"Setup completed in {int(time.time()) - start} seconds")
274
275     def assert_table(self, table_name):
276         assert spark.sql(f"SHOW TABLES IN {self.catalog}.{self.db_name}") \
277             .filter(f"isTemporary == false and tableName == '{table_name}') \
278             .count() == 1, f"The table {table_name} is missing"
```


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```
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275
276 def assert_table(self, table_name):
277     assert spark.sql(f"SHOW TABLES IN {self.catalog}.{self.db_name}") \
278         .filter(f"isTemporary == false and tableName == '{table_name}'") \
279         .count() == 1, f"The table {table_name} is missing"
280     print(f"Found {table_name} table in {self.catalog}.{self.db_name}: Success")
281
282 def validate(self):
283     import time
284     start = int(time.time())
285     print(f"\nStarting setup validation ...")
286     assert spark.sql(f"SHOW DATABASES IN {self.catalog}") \
287         .filter(f"databaseName == '{self.db_name}'") \
288         .count() == 1, f"The database '{self.catalog}.{self.db_name}' is missing"
289     print(f"Found database {self.catalog}.{self.db_name}: Success")
290     self.assert_table("registered_users_bz")
291     self.assert_table("gym_logins_bz")
292     self.assert_table("kafka_multiplex_bz")
293     self.assert_table("users")
294     self.assert_table("gym_logs")
295     self.assert_table("user_profile")
296     self.assert_table("heart_rate")
297     self.assert_table("workouts")
298     self.assert_table("completed_workouts")
299     self.assert_table("workout_bpm")
300     self.assert_table("user_bins")
301     self.assert_table("date_logs")
```

02-setup - Databricks

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Microsoft Azure databricks

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sbit-ws

prashant@scholarnest.com

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```
def assert_table(self, table_name):
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02-setup - Databricks

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    self.assert_table("user_profile")
    self.assert_table("heart_rate")
    self.assert_table("workouts")
    self.assert_table("completed_workouts")
    self.assert_table("workout_bpm")
    self.assert_table("user_bins")
    self.assert_table("date_lookup")
    self.assert_table("workout_bpm_summary")
    self.assert_table("gym_summary")
    print(f"Setup validation completed in {int(time.time()) - start} seconds")

def cleanup(self):
```


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```
300 self.assert_table("date_lookup")
301 self.assert_table("workout_bpm_summary")
302 self.assert_table("gym_summary")
303 print(f"Setup validation completed in {int(time.time()) - start} seconds")
304
305 def cleanup(self):
306     if spark.sql(f"SHOW DATABASES IN {self.catalog}").filter(f"databaseName == '{self.db_name}'").count() == 1:
307         print(f"Dropping the database {self.catalog}.{self.db_name}...", end='')
308         spark.sql(f"DROP DATABASE {self.catalog}.{self.db_name} CASCADE")
309         print("Done")
310     print(f"Deleting {self.landing_zone}...", end='')
311     dbutils.fs.rm(self.landing_zone, True)
312     print("Done")
313     print(f"Deleting {self.checkpoint_base}...", end='')
314     dbutils.fs.rm(self.checkpoint_base, True)
315     print("Done")
```

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Shift+Ctrl+Enter to run selected text

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Schedule

Share

Cmd 1

```
1 %run /Repos/SBIT/SBIT/02-setup
```

Command took 1.33 seconds -- by prashant@scholarnest.com at 8/22/2023, 10:36:10 PM on Prashant Pandey's Cluster

Cmd 2

```
1 SH = SetupHelper("dev")
2 SH.cleanup()
```

▶ (4) Spark Jobs

```
Deleting abfss://external@sbitsa.dfs.core.windows.net/data/raw...Done
Deleting abfss://external@sbitsa.dfs.core.windows.net/chkpt/checkpoints...Done
```

Command took 20.75 seconds -- by prashant@scholarnest.com at 8/22/2023, 10:37:19 PM on Prashant Pandey's Cluster

Cmd 3

```
1
```

Python ▶ ▾ - ✕

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Interrupt

Prashant Pandey's Clus...

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1 SH.setup()

Cancel ** Running command...

(36) Spark Jobs

Starting setup ...

Creating the database dev.sbit_db...Done

Creating registered_users_bz table...Done

Creating gym_logins_bz table...Done

Creating kafka_multiplex_bz table...Done

Creating users table...Done

Creating gym_logs table...Done

Creating user_profile table...Done

Creating heart_rate table...Done

Creating workouts table...Done

Creating completed_workouts table...Done

Creating workout_bpm table...

Cmd 4

Python

1

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Shift+Ctrl+Enter to run selected text

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Run all

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1 SH.validate()

(30) Spark Jobs

Starting setup validation ...

Found database dev.sbit_db: Success

Found registered_users_bz table in dev.sbit_db: Success

Found gym_logins_bz table in dev.sbit_db: Success

Found kafka_multiplex_bz table in dev.sbit_db: Success

Found users table in dev.sbit_db: Success

Found gym_logs table in dev.sbit_db: Success

Found user_profile table in dev.sbit_db: Success

Found heart_rate table in dev.sbit_db: Success

Found workouts table in dev.sbit_db: Success

Found completed_workouts table in dev.sbit_db: Success

Found workout_bpm table in dev.sbit_db: Success

Found user_bins table in dev.sbit_db: Success

Found date_lookup table in dev.sbit_db: Success

Found workout_bpm_summary table in dev.sbit_db: Success

Found gym_summary table in dev.sbit_db: Success

Setup validation completed in 4 seconds

Command took 4.06 seconds -- by prashant@scholarnest.com at 8/22/2023, 10:40:25 PM on Prashant Pandey's Cluster

Cmd 5

Python